

A Momentum Tilt Between Equities and Treasuries

thesispaper example

September 18, 2026

Abstract

We test whether tilting a static 60/40 SPY/TLT portfolio toward whichever asset has the higher trailing six-month return improves risk-adjusted performance after transaction costs. The example uses synthetic prices so that it runs offline.

Disclaimer. This document is research software output, not investment advice, and not evidence of future returns.

Contents

1	Thesis and research design	1
1.1	Hypotheses	2
2	Portfolio accounting	2
2.1	Currency conversion	2
2.2	NAV and target weights	2
3	Returns, wealth, and performance	3
4	Estimation and uncertainty	3
4.1	Gaussian maximum likelihood	3
4.2	Stationary bootstrap for the Sharpe ratio	4
5	Position sizing	4
6	Momentum tilt between equities and bonds	5
7	Results	5
8	Evidence and limitations	8

1 Thesis and research design

This note formalises an investment thesis as a testable mathematical object and evaluates it on historical data. The pipeline is deliberately mechanical: a strategy maps the price history available at time t to target weights (Section 2); a backtest converts those weights into a net-of-cost return series; and every number quoted in the text is generated from that series, not typed by hand (Sections 3–5 define the estimators; Section 7 reports them).

The research sample contains 2608 sessions from 2015-01-01 to 2024-12-31. Trading costs are charged at 5 basis points per unit of turnover. These assumptions are fixed before the results are read, and they determine what the figures can and cannot show.

The key design rule is *no look-ahead*: the weight vector applied to the return of session t may depend only on information dated strictly before that return is realised. Formally, if $\mathcal{F}_{t-1}$ is the information set at the close of session $t-1$, then

$$w_t = g(\mathcal{F}_{t-1}), \quad r_t^{\text{strat}} = \sum_i w_{i,t} r_{i,t} - c \sum_i |w_{i,t} - w_{i,t}^{\text{drift}}|, \quad (1)$$

where $w_{i,t}^{\text{drift}}$ is the previous target weight after drifting with the market and $c = \text{CostBps}/10^4$ is the proportional cost rate.

1.1 Hypotheses

H1 The momentum-tilted portfolio has a higher Sharpe ratio than static 60/40.

H2 The improvement survives proportional costs of 5 basis points.

2 Portfolio accounting

2.1 Currency conversion

Let asset i be quoted in currency Q_i and let the account base currency be B . If X_t^{B,Q_i} denotes units of Q_i per one unit of B , a native price $P_{i,t}^{Q_i}$ has base-currency value

$$P_{i,t}^B = \frac{P_{i,t}^{Q_i}}{X_t^{B,Q_i}}. \quad (2)$$

Equivalently, with a direct multiplier $Y_t^{Q_i,B} = (X_t^{B,Q_i})^{-1}$ in units of B per unit of Q_i ,

$$P_{i,t}^B = P_{i,t}^{Q_i} Y_t^{Q_i,B}. \quad (3)$$

Forward-filling an FX series onto an asset's trading dates is an alignment convention, not an estimate of an unobserved intraday rate. When all assets share the base currency, both equations are the identity.

2.2 NAV and target weights

With cash C_t , position quantity $q_{i,t}$, and base-currency price $P_{i,t}^B$, net asset value (NAV) is

$$V_t = C_t + \sum_{i=1}^m q_{i,t} P_{i,t}^B. \quad (4)$$

For target weights $w_i \geq 0$ satisfying $\sum_i w_i \leq 1$, the ideal target quantity is

$$q_{i,t}^* = \frac{w_i V_t}{P_{i,t}^B}. \quad (5)$$

Equation (5) is the bridge between economic allocation and position size. Unallocated weight, $1 - \sum_i w_i$, remains in cash and earns zero. The backtest uses fractional quantities so that rounding does not obscure the mathematics.

3 Returns, wealth, and performance

For a consistently adjusted price series P_t , the one-period simple return is

$$r_t = \frac{P_t}{P_{t-1}} - 1, \quad (6)$$

and the log return is

$$\ell_t = \log\left(\frac{P_t}{P_{t-1}}\right) = \log(1 + r_t). \quad (7)$$

Simple returns aggregate across simultaneously held assets; log returns add through time.

Starting from wealth V_0 , compounded wealth after T returns is

$$V_T = V_0 \prod_{t=1}^T (1 + r_t). \quad (8)$$

Total return and compound annual growth rate over Y calendar years are

$$R_{\text{tot}} = \frac{V_T}{V_0} - 1, \quad \text{CAGR} = (1 + R_{\text{tot}})^{1/Y} - 1. \quad (9)$$

Using $A = 252$ sessions per year and a zero risk-free rate, annualised volatility and the Sharpe ratio are

$$\hat{\sigma}_{\text{ann}} = \hat{\sigma}_r \sqrt{A}, \quad \hat{S} = \frac{\bar{r}}{\hat{\sigma}_r} \sqrt{A}. \quad (10)$$

For the rolling graph, $\bar{r}$ and $\hat{\sigma}_r$ are recomputed over a trailing window using the sample standard deviation; the estimate is path-dependent and moves sharply when an extreme observation enters or leaves the window.

Let $M_t = \max_{0 \leq s \leq t} V_s$ be the running high-water mark. Drawdown and maximum drawdown are

$$D_t = \frac{V_t}{M_t} - 1, \quad D_{\text{max}} = \min_{0 \leq t \leq T} D_t. \quad (11)$$

These summaries describe one realised path. They do not quantify parameter uncertainty, multiple testing, tail severity beyond the observed sample, or the probability of future loss.

4 Estimation and uncertainty

4.1 Gaussian maximum likelihood

For returns $x_1, \dots, x_n$, assume as a working approximation that $x_j \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. The log-likelihood, up to an additive constant, is

$$\mathcal{L}(\mu, \sigma^2) = -\frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^n (x_j - \mu)^2. \quad (12)$$

Maximisation yields

$$\hat{\mu} = \frac{1}{n} \sum_{j=1}^n x_j, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{j=1}^n (x_j - \hat{\mu})^2. \quad (13)$$

The divisor n is the maximum-likelihood convention, not the unbiased one. The estimated standard error of the mean is

$$\widehat{\text{se}}(\widehat{\mu}) = \frac{\widehat{\sigma}}{\sqrt{n}}. \quad (14)$$

For the daily strategy returns of this study, $\widehat{\mu} = 1.083e-04$, $\widehat{\sigma}^2 = 7.346e-05$ and $\widehat{\text{se}}(\widehat{\mu}) = 1.678e-04$. An estimated edge is only interesting relative to this error: the statistic $\widehat{\mu}/\widehat{\text{se}}(\widehat{\mu})$ is a t -type ratio, not a significance test, because returns are neither established as Gaussian nor as independent.

4.2 Stationary bootstrap for the Sharpe ratio

Daily returns are serially dependent enough that the iid interval is unreliable. The stationary bootstrap of Politis and Romano resamples blocks with geometric random length. With mean block length $1/p$, define the index sequence recursively: I_1 is uniform on $\{1, \dots, n\}$, and

$$I_{k+1} = \begin{cases} (I_k \bmod n) + 1, & \text{with probability } 1 - p, \\ \text{uniform on } \{1, \dots, n\}, & \text{with probability } p. \end{cases} \quad (15)$$

For replication $b = 1, \dots, B$ (here $B = 1000$), form $x_k^{*b} = x_{I_k}$ for $k = 1, \dots, n$ and compute $\widehat{S}^{*b}$ from Equation (10). The percentile interval for the Sharpe ratio is

$$\left[\widehat{S}_{(\alpha/2)}^*, \widehat{S}_{(1-\alpha/2)}^* \right], \quad (16)$$

the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of the replications. For this study the interval is $[-0.35, 0.82]$ around the point estimate 0.20. The interval captures sampling noise under the resampling assumptions only; it does not correct for choosing the strategy after seeing data.

5 Position sizing

For a small risky return R and fraction f of wealth invested, a second-order expansion of expected log growth is

$$\mathbb{E}[\log(1 + fR)] \approx f \mathbb{E}[R] - \frac{f^2}{2} \mathbb{E}[R^2]. \quad (17)$$

When the mean is small relative to volatility, $\mathbb{E}[R^2] \approx \text{Var}(R)$, and maximising over f gives the familiar approximate optimiser

$$f^* \approx \frac{\mathbb{E}[R]}{\text{Var}(R)}. \quad (18)$$

Because $\mathbb{E}[R]$ and $\text{Var}(R)$ are estimated with error (Section 4), the full-Kelly fraction is systematically too aggressive. A fractional-Kelly rule scales it by $\lambda \in (0, 1]$ and imposes a cap $f_{\max}$, after subtracting the per-trade cost c :

$$f = \min \left(f_{\max}, \lambda \frac{\max(\widehat{\mu} - c, 0)}{\widehat{\sigma}^2} \right). \quad (19)$$

This is “Kelly-style”, not an exact Kelly solution: return distributions are not established, parameters are estimated, and the resulting weights are then constrained to $\sum_i w_i \leq 1$ as in Section 2. Growth is concave in f around f^* , so a fraction $\lambda < 1$ gives up a small amount of expected growth in exchange for a substantially smaller variance and drawdown.

For this study, plugging the estimates of Section 4 into Equation (18) gives an approximate full-Kelly fraction of 1.47; with $\lambda = 0.50$ and cap $f_{\max} = 1.00$, Equation (19) gives 0.74. A fraction above one would imply leverage; the strategy does not use these figures, which serve only to show how fragile the full-Kelly value is given the standard error above.

6 Momentum tilt between equities and bonds

Let P_t^E and P_t^B be the adjusted closes of SPY (equities) and TLT (bonds), and let L be the lookback (126 sessions). The trailing momentum of each asset at the close of session t is

$$m_t^a = \frac{P_t^a}{P_{t-L}^a} - 1, \quad a \in \{E, B\}. \quad (20)$$

The leader is $\ell_t = \arg \max_a m_t^a$. With base equity weight $w_0 = 0.6$ and tilt $\tau = 0.2$, the target weights are

$$w_t^{\ell_t} = w_0^{\ell_t} + \tau, \quad w_t^a = w_0^a - \tau \quad (a \neq \ell_t), \quad \sum_a w_t^a = 1. \quad (21)$$

These use only $\mathcal{F}_t$ at the close of session t and are applied to returns from $t + 1$, consistent with Equation (1). The excess return over the static 60/40 reference portfolio is linear in the tilt:

$$r_{t+1}^{\text{strat}} - r_{t+1}^{\text{static}} = \tau s_t (r_{t+1}^E - r_{t+1}^B), \quad s_t = \begin{cases} +1, & \ell_t = E, \\ -1, & \ell_t = B. \end{cases} \quad (22)$$

Because the cost is proportional to turnover, the net excess return per period has expectation

$$\mathbb{E}[r_{t+1}^{\text{strat}} - r_{t+1}^{\text{static}}] = \tau \mathbb{E}[s_t (r_{t+1}^E - r_{t+1}^B)] - c \mathbb{E}[\text{TO}_t], \quad (23)$$

where TO_t is turnover. A tilt only trades when the leader flips, and each flip moves 2τ of weight in total. Hypothesis H2 (edge survives costs) is the statement that Equation (23) is positive at the assumed c ; H1 concerns the Sharpe ratio of the same series and is evaluated empirically in Section 7.

7 Results

All values below are generated from the backtest output; none is typed by hand. The sample spans 2608 sessions from 2015-01-01 to 2024-12-31 with costs of 5 basis points.

Table 1: Performance summary (net of costs). The interval for the strategy Sharpe ratio is the stationary-bootstrap percentile interval of Equation (16).

	Strategy	Benchmark
CAGR	1.82%	-3.00%
Sharpe ratio	0.20	-0.08
Maximum drawdown	-35.33%	-50.27%
Total return	20.54%	
Annualised volatility	13.61%	
Sharpe 95% interval	[-0.35, 0.82]	

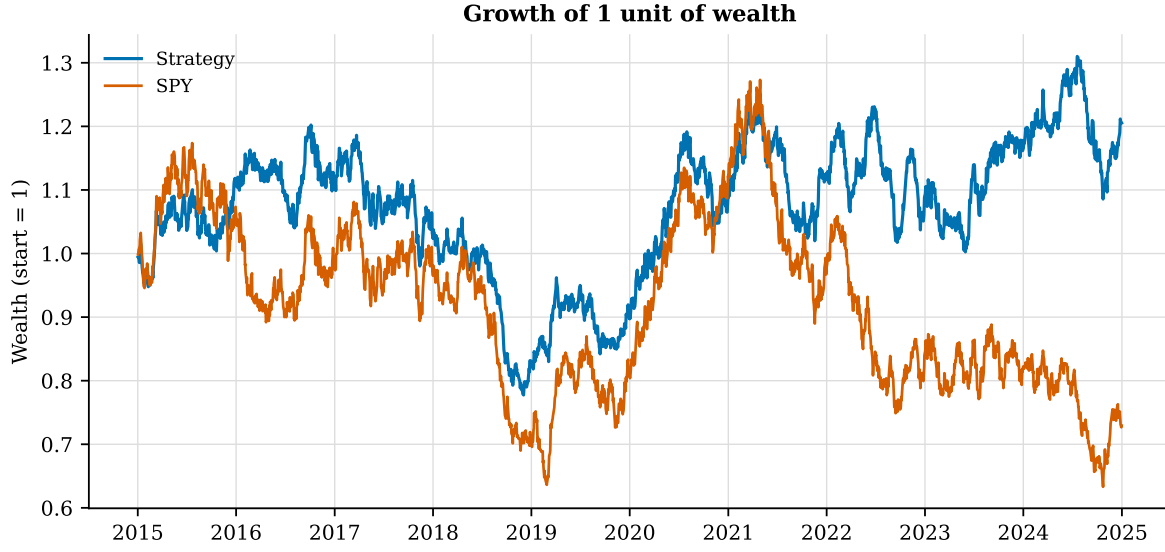


Figure 1: Cumulative wealth of the strategy and its benchmark (Equation (8)).

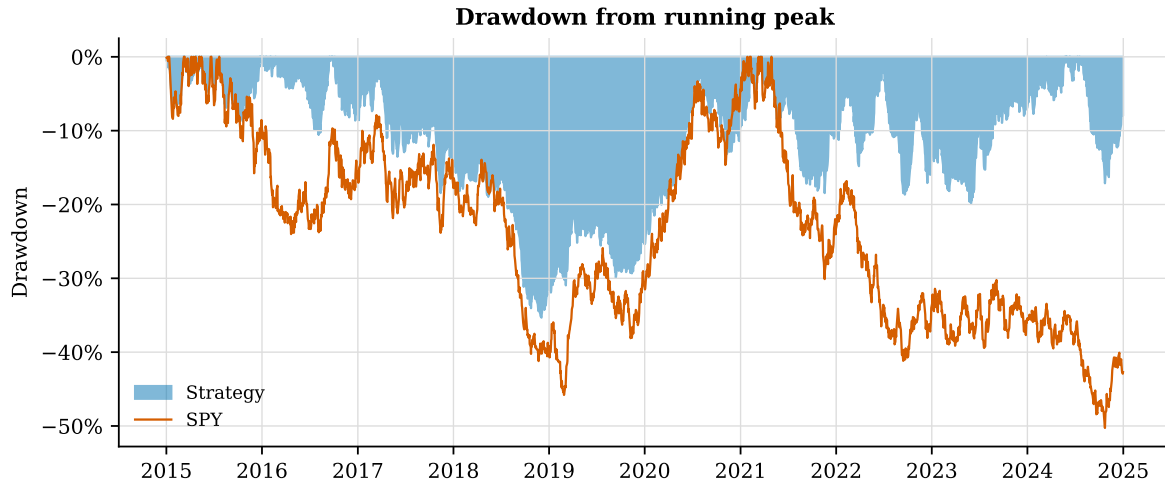


Figure 2: Drawdown from the running peak (Equation (11)).

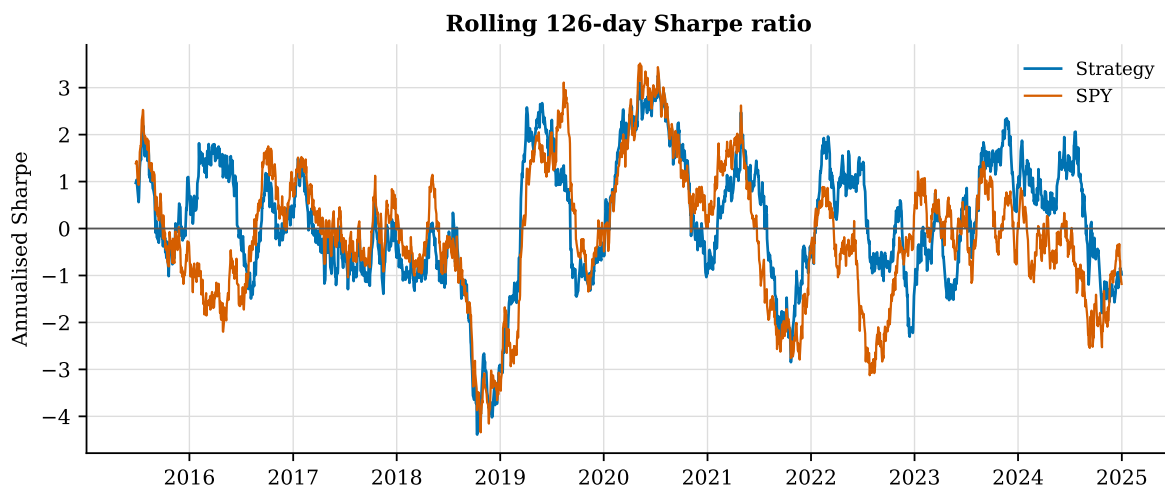


Figure 3: Rolling annualised Sharpe ratio (Equation (10)). Adjacent windows overlap almost completely, so the path is descriptive rather than a sample of independent estimates.

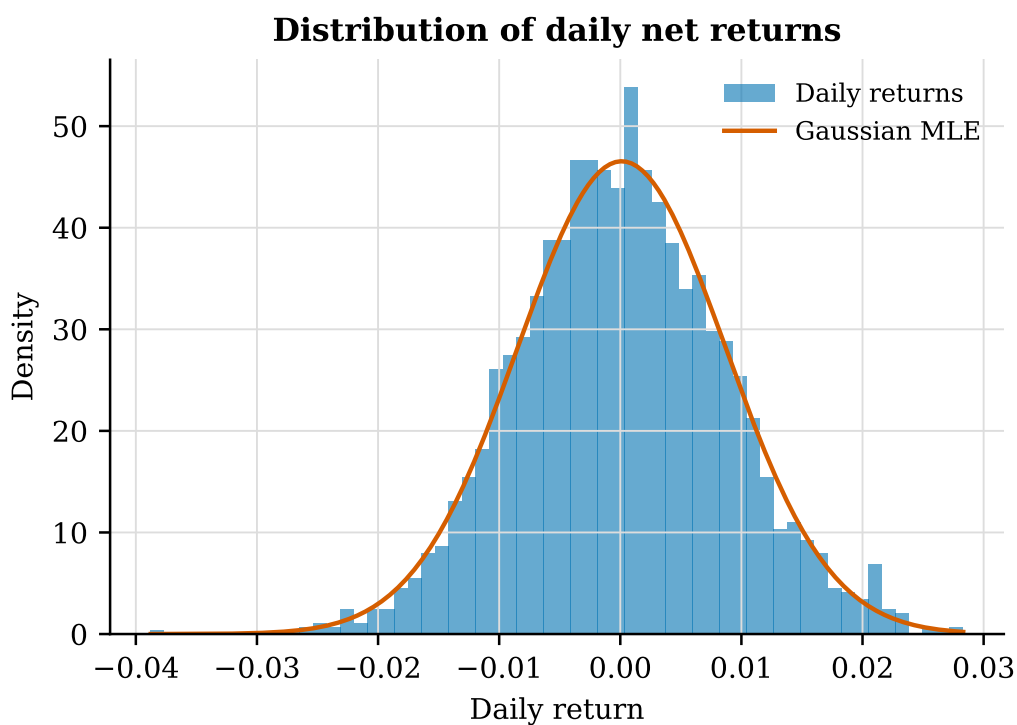


Figure 4: Distribution of daily strategy returns. Fat tails and skew, if visible, contradict the Gaussian working assumption of Equation (12).

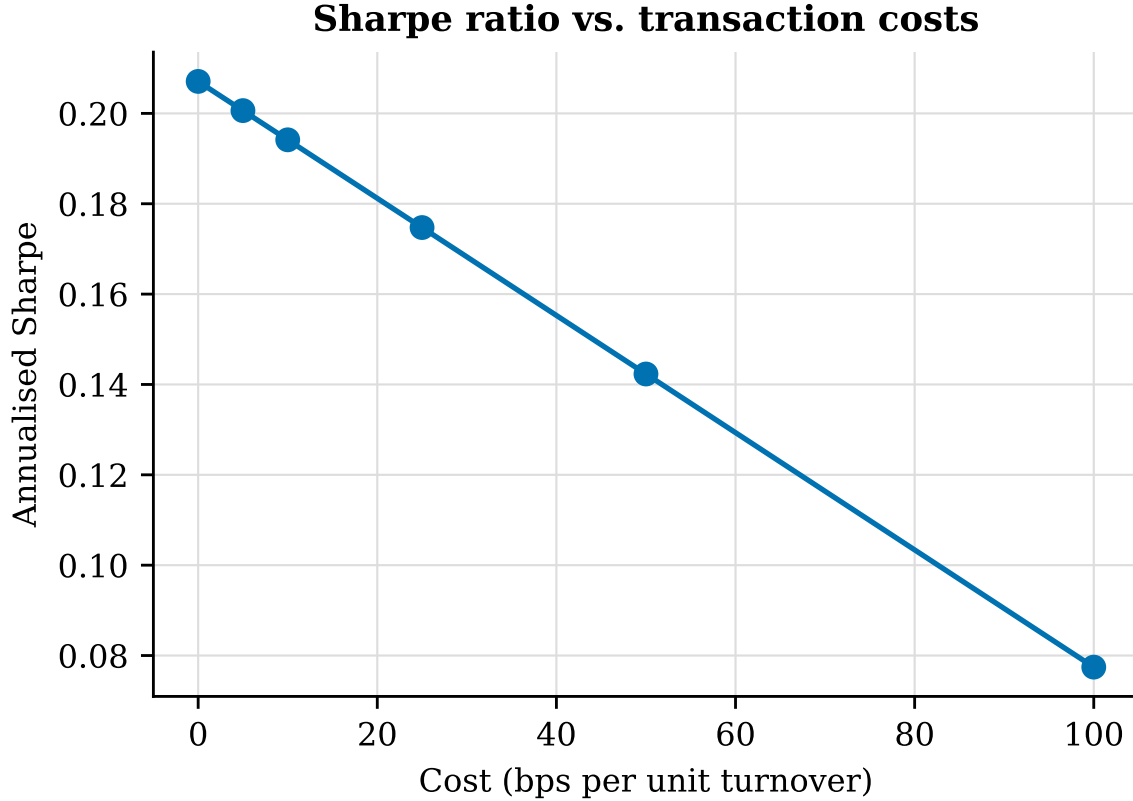


Figure 5: Sensitivity of performance to the transaction-cost assumption. A result that only survives at low cost is fragile.

8 Evidence and limitations

The figures show one realised path and are not evidence of dominance. The following caveats apply to every paper produced by this pipeline, whatever the numbers say.

- **Sample length.** The sample has 2608 sessions (2015-01-01 to 2024-12-31). The standard error of a Sharpe ratio estimated from T years of data is roughly $\sqrt{(1 + S^2/2)/T}$, so short samples cannot distinguish skill from luck; see the interval $[-0.35, 0.82]$.
- **Costs.** Costs are a proportional 5 basis points per unit of turnover. Real costs include spreads that widen in stress, market impact, commissions, financing, and taxes, none of which are modelled.
- **Overfitting and multiple testing.** The thesis was formulated with knowledge of history. Every parameter choice, ticker choice, and discarded variant increases the chance that the best-looking result is noise. The bootstrap interval does not account for this selection.
- **Model assumptions.** Gaussian and iid assumptions are working approximations. Financial returns are heteroskedastic, skewed, and heavy-tailed.
- **Data.** Adjusted closes are assumed correct and free of survivorship bias; the risk-free rate is taken as zero, and rankings can change with other choices.

- **No out-of-sample test.** Unless stated otherwise, no untouched holdout period was reserved.

This document is not investment advice. It is an automatically generated research note; nothing in it has been validated for live capital, and past performance says little about future returns.

Conclusion

The chain of argument is mechanical: consistent prices give returns (Equation (6)); weights that use only past information give a net-of-cost strategy return (Equation (1)); compounding gives wealth, growth, and drawdown (Equations (8)–(11)); and estimation theory gives an honest statement of uncertainty (Equation (16)). Over 2015-01-01 to 2024-12-31 the strategy delivered a CAGR of 1.82% and a Sharpe ratio of 0.20 against -3.00% and -0.08 for the benchmark, with a maximum drawdown of -35.33%. Given the limitations of Section 8, these are hypotheses for further out-of-sample study, not conclusions about the future.